Lab 4 session

Harris Corner Detector:

Basic steps for applying the Harris corner detector:

1. Smoothing of the selected gray image using a Gaussian filter.

Sigma = 1, size of [5, 5]. See notes file (table 1)

2. Calculate the first partial derivatives (first order) of the smoothed image with respect to x and y (the horizontal and vertical components of the gradient) using Prewit filters and the convolution product. See notes file (table 2)
3. Calculate the second partial derivatives (of second order) of the previous form with respect to x, y, and x & y altogether using equations (1, 2 and 3 in notes file).
4. Calculate the second moment matrix components (M_{11}, M_{12}, M_{21}, M_{22}) using another Gaussian filter (w) and convolution process. Sigma = 2, size of [13, 13]. See notes file (table 3)
5. Find interest points using the Harris criterion for important points/ feature:

$$R = \frac{\det M}{trace\ M}$$

det (M) = M_{11} . M_{22} – M_{12} . M_{21}
trace (M) = M_{11} + M_{22}
Make sure that all NaN values become zeros.

6. Apply the non-maximum suppression process using a window of the size [10, 10] for example sliding overall R image, by suppress the central pixel if its value is less than the local maximum value of the window.
7. Normalize the result to become in the range [0, 1].
8. Apply thresholding process to the normalized image using some values by trial and error (just to eliminate some extra unnecessary corners).

9. Draw a bounding box or a circle around the corners.

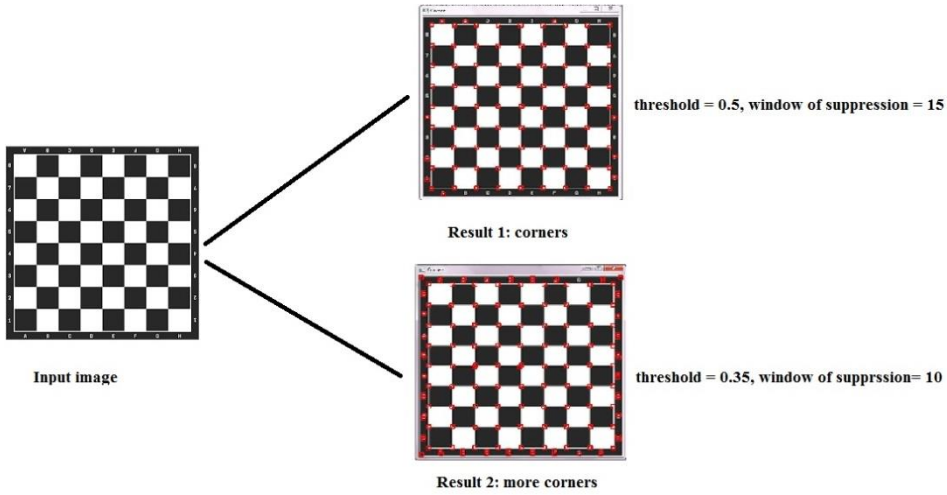


Fig. (1) The final results of applying Harris detector on a Chess board image